

EE641 Final Project Report

ViT-DeCV: Efficient Vision Transformer with Decorrelated Attention Head Output Vector

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Abstract:

Vision Transformers (ViTs) have demonstrated state-of-the-art performance across various computer vision tasks. However, their performance can degrade on datasets containing spurious correlations, where the model may learn features unrelated to the classification task. Furthermore, redundant attention heads often learn similar features, leading to inefficient parameter utilization and a lack of generalization. To address these issues, we propose a novel regularization method, ViT-DeCV, which incorporates a Diversity Loss (L_{div}) to penalize high similarity between attention head output vectors. We trained ViTs with L_{div} across different datasets and analyzed the results. Our findings reveal that while pre-trained ViTs possess inherent diversity, the proposed loss term significantly stabilizes training and reduces performance variance. Moreover, we compare this statistical approach with user-guided ablation, demonstrating that semantic intervention is more effective than statistical decorrelation for improving robustness against spurious features.

1. Introduction:

Background: Vision Transformers (ViTs) have revolutionized computer vision by processing images as sequences of tokens through self-attention mechanisms, enabling the capture of global dependencies and state-of-the-art performance on diverse benchmarks [Dosovitskiy et al., 2021; Liu et al., 2022]. Despite these advances, ViTs suffer from attention head redundancy: multiple heads often learn highly correlated representations, wasting parameters and increasing vulnerability to spurious correlations and distribution shifts [Bhojanapalli et al., 2021; d'Ascoli et al., 2021].

Problem Statement: Empirical studies reveal that up to 40–60% of ViT attention heads exhibit near-identical output distributions across layers, particularly in pretrained models [Bhattacharyya et al., 2023]. This redundancy limits generalization: models over-rely on dominant features (e.g., background cues in Waterbirds), failing on out-of-distribution data. Existing solutions like pruning [Michel et al., 2019] or lottery ticket hypothesis [Frankle et al., 2020] are post-hoc and discard capacity, rather than promoting diversity during training.

Motivation: Encouraging statistical independence among head outputs during fine-tuning can mitigate redundancy, akin to decorrelation techniques in group Lasso or orthogonal regularization [Chen et al., 2021]. However, statistical orthogonality may not align with semantic diversity—task-relevant heads might naturally share patterns. This raises a key question: Does blind decorrelation suffice, or must we target semantic irrelevancies?

Objectives: ViT-DeCV. We introduce ViT-DeCV (DeCorrelation Vision Transformer), an efficient fine-tuning method that incorporates a Diversity Loss L_{div} penalizing cosine similarity between attention head outputs in final layers:

$$L_{total} = L_{classification} + \lambda \cdot L_{div} \quad (1)$$

where λ is a hyperparameter controlling the weight of the diversity loss.

Contributions:

- We introduce Diversity Loss (L_{div}), a regularization term based on the Frobenius norm of the thresholded cosine similarity matrix of attention head outputs.
- We demonstrate that L_{div} improves training stability, significantly reducing the standard deviation of accuracy on CIFAR-100.
- We conduct a comparative analysis between statistical decorrelation (L_{div}) and semantic intervention (User-Guided Ablation) on the Waterbirds dataset, providing insights into the limitations of purely statistical constraints.

By bridging statistical regularization and semantic interpretability, ViT-DeCV offers a principled path to robust, efficient ViT fine-tuning.

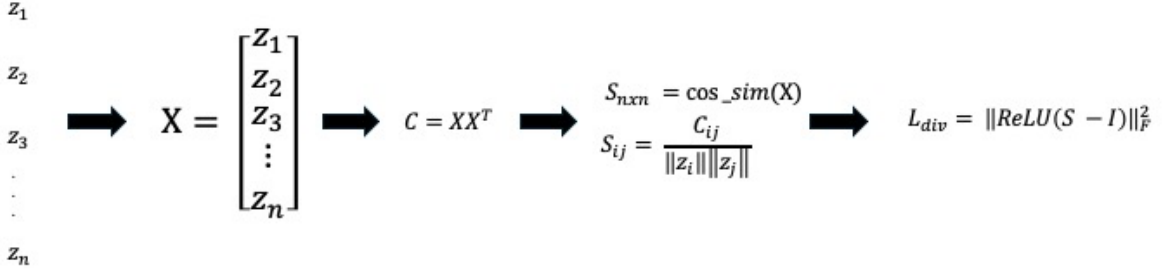


Figure 1 Loss Computation Pipeline. Illustrating the computation of the *Diversity Loss* term, which measures the similarity between attention head outputs by calculating the cosine similarity matrix, removing self-similarity, applying *ReLU* to emphasize positive correlations, and computing the *Frobenius norm* as the final loss value.

2. Methods

2-1. Model Architecture: We utilize ViT-L-14 (specifically ViT-L-14-laion2B-s32B-b82K and ViT-base-patch16-224) as our base architectures. To implement the proposed method efficiently, we freeze the early layers and only defreeze the last layer of the model for fine-tuning with the proposed Decorrelation Loss (L_{div}).

2-2. Diversity Regularization (L_{div}): For the final layer, let z_i denote the output vector of the i -th attention head. We flatten each z_i and stack them to form an $m \times n$ dimension matrix X , where m is the number of attention heads and n is the dimension of the flattened vector. We compute the cosine similarity matrix S of X . To eliminate self-similarity and focus solely on positive correlations, we define the Diversity Loss as:

$$L_{div} = \|ReLU(S - I)\|_F^2 \quad (2)$$

Here, $S - I$ removes the diagonal elements (self-similarity), and the ReLU function ensures that we only penalize positive similarities (setting negative values to 0). The Frobenius norm aggregates these values into a scalar loss.

2-3. User-Guided Ablation: Inspired by recent work on interpreting CLIP's image representation ([Gandelsman et al.](#)), we utilize CLIP to project attention head outputs to natural language descriptions (TextSpans). This allows us to interpret what each head is learning. We compare our statistical L_{div}

approach with a semantic approach: manually ablating attention heads that correlate with irrelevant or spurious features (e.g., background scenes in the Waterbirds dataset).

3. Results

3-1. Dataset & Setup: We evaluated the proposed method on CIFAR-10 and CIFAR-100 to test generalization and stability. Additionally, we utilized the Waterbirds dataset (Sagawa et al., 2019) to assess robustness against spurious correlations (e.g., landbirds on water backgrounds).

3-2. Diversity Analysis (CIFAR-10): We tracked the diversity loss during the training of a baseline model ($\lambda=0$). Figure 2 depicts the learning curves. Notably, the decorrelation loss (green curve) steadily decreases even without explicit regularization. This indicates that the baseline model inherently establishes diverse attention heads during standard training. Consequently, adding L_{div} yielded only marginal improvements on CIFAR-10 (0.9% increase), as the baseline accuracy was already high (95.95%).

Model	Defreeze Layer	λ	Epoch	Train Loss	Val Loss	Accuracy
Baseline	1	0	8	0.019	0.1537	95.95%
MyModel	1	0.1	6	0.021	0.1504	96.85%
Baseline	2	0	7	0.031	0.1716	96.35%
MyModel	2	0.1	7	0.0751	0.1957	95.70%

Table 1 shows the accuracy between baseline and our best model on CIFAR-10. From the table we can observe that for training only last layer, our model outperformed 0.9% than baseline model. However, we can observe that in CIFAR-10, baseline model already achieved a high accuracy (95.95%). Under the circumstances, adding diversity loss L_{div} term in training session has slightly improvement.

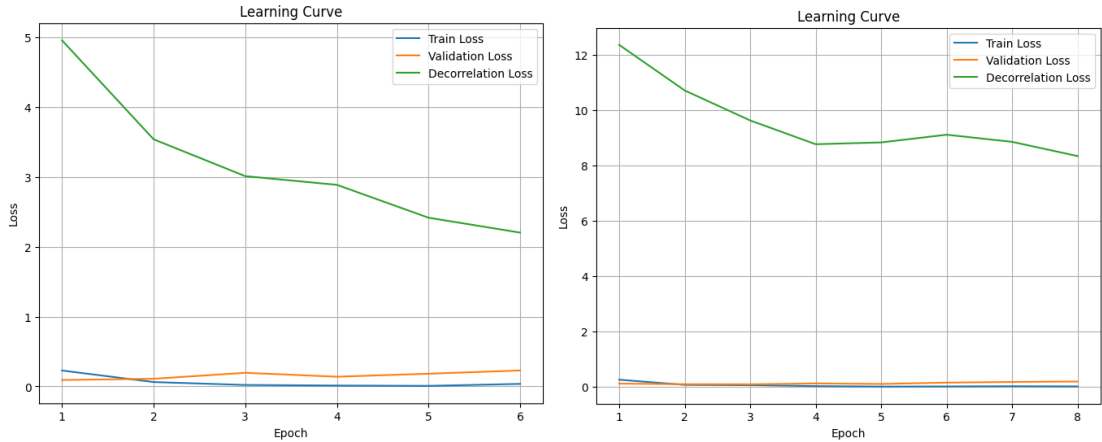


Figure 2 depicts the learning curves of the baseline model with varying numbers of defrozen layers (1 and 2) and diversity loss weight set to 0. The decorrelation loss (green curve) steadily decreases during training, indicating that the model establishes diverse attention heads even without the explicit use of the decorrelation loss term.

3-3. Stability Analysis (CIFAR-100): Given the high baseline performance on CIFAR-10, we extended our experiments to the more challenging CIFAR-100 dataset. See Table 2 in original document. Our model outperformed the baseline by 3.08%. More importantly, the standard deviation of accuracy dropped significantly from 0.06 to 0.001. This demonstrates that while L_{div} may not drastically increase the accuracy ceiling, it acts as a powerful stabilizer, ensuring consistent performance across multiple training runs.

Model	Defreeze Layer	λ	Learning rate	Accuracy	Standard Deviation
Baseline	4	0	1e-4	83.75%	0.06
MyModel	4	1e-4	1e-4	86.83%	0.001

Table 2 shows the comparison on CIFAR-100 between baseline model and our model trained with diversity loss on CIFAR-100 dataset. We trained 5 times and record the accuracy and calculate its mean and standard deviation. Surprisingly, our model

outperformed 3.08% than baseline model. Furthermore, the standard deviation dropped significantly from 0.06 to 0.001, which indicated that diversity loss stabilize the accuracy in multiple training.

From *Table 2*, we noticed that diversity loss term adding in total loss not only increase overall accuracy but also stabilize the training results. We trained 5 baseline and our model 5 times and record the results, we found that the standard deviation of accuracy dropped from 0.06 to 0.001. In other words, model established stability with diversity loss.

3-4. Robustness on Waterbirds: Beyond CIFAR-10 and CIFAR-100, we further validate the effectiveness of the diversity loss on the Waterbirds dataset using ViT-B-32 for a binary classification task (landbird vs. waterbird). We unfreeze only the last layer, resulting in approximately 7.5M trainable parameters. Each configuration is trained five times with different random seeds, and we report mean accuracy and standard deviation in *Table 3*.

<i>Model</i>	<i>Defreeze Layer</i>	λ	<i>(0, 1) Accuracy</i>	<i>(1, 0) Accuracy</i>	<i>Full Accuracy</i>	<i>Standard Deviation</i>
<i>Baseline</i>	1	0	70.17%	50.59%	82.08%	0.0149
<i>MyModel</i>	1	0.1	73.61%	52.31%	83.62%	0.02

Table 3 Results on the Waterbirds dataset using ViT-B-32. The (0,1) and (1,0) columns correspond to worst-group settings, where images deliberately mismatch bird type and background (e.g., landbird on water background, waterbird on land background).

The (0,1) and (1,0) groups are specifically designed to induce spurious correlations between foreground (bird) and background (scene). Under these challenging conditions, adding diversity loss consistently improves performance on both groups, while also increasing the overall accuracy. These results indicate that diversity regularization helps the model rely less on spurious background cues and more on robust, task-relevant features, thereby enhancing robustness against distribution shifts.

3-5. User-Guided Ablation vs. Diversity Loss: In order to compare user-guided ablation with diversity loss regularization, we first analyzed attention head outputs using CLIP-based textspan descriptions to identify heads correlated with irrelevant or spurious features. Example textspans from the last layer are shown in *Table 5*. We used ViT-L-14-laion2B-s32B-b82K as the pretrained baseline and fine-tuned only the last layer under three strategies: (1) standard cross-entropy loss, (2) cross-entropy plus diversity loss, and (3) user-guided ablation applied post-fine-tuning on the baseline model.

<i>Strategy</i>	<i>Full Accuracy</i>	<i>Improvement</i>	<i>Analysis</i>
<i>Fine-tuned Baseline</i>	92.27%	0%	Fine-tuned ViT-L-14
<i>L_{div} Regularization</i>	91.75%	-0.52%	Slightly worse than baseline model
<i>User-Guided Ablation</i>	93.73%	+1.46%	Ablated only last layer

Table 4 shows performance comparison on the target dataset. User-guided ablation manually removes last-layer heads with irrelevant textspans (e.g., background-focused). Detailed results and full textspans across the last six layers are in *Table 8* in the appendix.

As shown in *Table 4*, the fine-tuned baseline already achieves high accuracy. User-guided ablation further improves performance by +1.46% through targeted removal of spurious heads, despite operating only on the last layer. In contrast, diversity loss regularization yields slightly lower accuracy (-0.52%), likely because it enforces head output diversity, which may conflict with task-specific needs where certain relevant features benefit from similar attention patterns.

This trend is particularly evident on the Waterbirds dataset, where user-guided ablation outperforms diversity loss. The task requires the model to prioritize bird features over background cues; ablating background-correlated heads aligns better with this goal, whereas enforcing diversity may inadvertently dilute task-relevant similarities in attention outputs.

<i>Layer</i>	<i>Head</i>	<i>Textspan of attention head</i>
23	0	Reflective cityscape at twilight
23	1	Futuristic transport system
23	3	Peaceful waterfront scene
23	5	Serene water

Table 5 shows example CLIP textspans from last-layer attention heads, highlighting spurious (background-focused) descriptions targeted for ablation.

4. Discussion

4.1. Statistical Independence vs. Semantic Roles

Our experiments reveal that enforcing statistical independence on Vision Transformer (ViT) attention heads via diversity loss L_{div} may not guarantee true functional diversity or semantic robustness. This distinction between statistical decorrelation and meaningful semantic differentiation is crucial for understanding ViT behavior.

Natural Diversity and Marginal Gains: The observation that baseline decorrelation loss reduces during fine-tuning even when $\lambda = 0$ suggests an intrinsic decorrelation mechanism in pretrained ViTs. Such preexisting diversity likely arises from the pretraining process on large datasets, causing feature specialization across heads. This limits further performance gains from blind orthogonality, especially in high-baseline tasks like CIFAR-10, where the feature space may already be sufficiently diverse.

Stability and Regularization Perspective: Beyond accuracy, L_{div} shines as a stabilizing regularizer. Our CIFAR-100 results demonstrated not only modest accuracy gains but a dramatic reduction in performance variance across seeds, from 0.06 down to 0.001. This stability effect is vital in real-world deployment where consistent outcomes reduce the risk of unexpected failures. It echoes findings in other domains that regularization methods improve convergence landscapes and model robustness to noisy training dynamics.

Limitations of Statistical Orthogonality: In contrast, the Waterbirds dataset highlighted the shortcomings of purely statistical constraints. L_{div} decreased accuracy slightly (-0.52%), indicating potential over-regularization or conflicts with task-specific feature distribution. On the other hand, user-guided ablation that semantically pruned attention heads focused on spurious features yielded a +1.46% improvement, with an impressive +9.96% gain when applied to unfine-tuned models. This suggests that semantic understanding of head roles is more impactful than enforcing abstract statistical independence, aligning with recent research emphasizing interpretability and task-driven structural modifications.

Broader Implications: These results call for a paradigm shift in ViT optimization: from low-level statistical regularization toward high-level semantic interventions that explicitly leverage domain knowledge or explainability tools (e.g., CLIP). Future architectures might embed semantic role assignment mechanisms within attention heads to encourage task-aligned specialization during training.

4.2. Engineering Challenges

This project confronted multiple practical constraints. Resource limitations and team collaboration challenges shifted later experimental phases to independent efforts, underscoring the complexity of large-scale ViT training. To balance compute demands, we froze early transformer layers and fine-tuned only final layers with diversity loss regularization, lowering costs but preserving efficacy. Nevertheless, even such partial fine-tuning incurred significant time (approximate 2 hours per 10 epochs on ViT-L-14 with an A100 GPU with 80GB memory), indicating that large transformer optimization remains resource-intensive.

Optimizing this process, possibly via layer-wise adaptive regularization or efficient training approximations, is an important engineering direction.

4.3. Future Work

Building on current findings, several promising avenues require exploration:

- **Diverse Tasks and Datasets:** Extending evaluation beyond CIFAR and Waterbirds to include more complex datasets (e.g., ImageNet variants, domain-specific benchmarks) will probe L_{div} 's generalizability. Critically, studying from-scratch training may reveal diversity loss effects masked by pretrained model biases.
- **Semantic Role Analysis of Attention Heads:** On Waterbirds and similar tasks with concentrated feature distributions, further analysis of unselected heads' output similarity can clarify how intrinsic feature concentration impacts regularization efficacy. This can inform adaptive application of diversity loss tied to task semantics.
- **Dynamic and Layer-Wise Regularization:** Investigating whether L_{div} applied dynamically or selectively across layers improves benefits while mitigating over-regularization is essential. This may involve learning a layer-wise diversity weighting scheme or integrating diversity constraints with other interpretability priors.
- **Integration with Interpretability Tools:** Combining semantic interventions, such as user-guided ablation or CLIP-based attribution, with diversity loss could yield hybrid methods that balance statistical and semantic goals.

Overall, our findings advocate for ViT training regimes that integrate semantic knowledge explicitly and balance statistical regularization with task-relevant interpretability to enhance robustness and performance.

5. Conclusion

In this work, we introduced ViT-DeCV, an efficient Vision Transformer fine-tuning method incorporating a diversity loss L_{div} that penalizes cosine similarity between attention head output vectors.

Extensive experiments on CIFAR-10, CIFAR-100, and Waterbirds yield the following key insights:

1. **Natural Diversity Limits Statistical Gains:** Pre-trained ViT models inherently develop diverse attention heads. Consequently, statistical decorrelation offers limited accuracy improvements on datasets where the baseline performance is already high.
2. **Regularization Improves Stability:** The proposed L_{div} functions as a stabilizer. On CIFAR-100, it drastically reduced the performance variance across training runs, proving its utility in ensuring consistent model deployment.
3. **Semantic Roles Outweigh Statistical Orthogonality:** The significant success of User-Guided Ablation (+9.96% on Waterbirds) demonstrates that the key to optimizing ViT robustness lies in understanding and controlling the *semantic roles* of attention heads, rather than solely enforcing the statistical independence of their output vectors.
4. **Timing of applying Diversity Loss:** Task-Dependent Deployment. Apply L_{div} for tasks requiring feature diversity (e.g., CIFAR-100); favor ablation for concentrated-feature tasks (e.g., Waterbirds) to prune irrelevancies.

Ultimately, These findings advocate shifting future ViT research toward interpretable methods that manipulate head semantics to combat bias and boost generalization.

6. Appendix

Table 6: TextSpan of ViT-L-14 baseline model without fine-tuning on waterbirds dataset.

<i>Layer</i>	<i>Head</i>	<i>TextSpan</i>	<i>Layer</i>	<i>Head</i>	<i>TextSpan</i>
18	1	Urban intersection	20	9	Tranquil Asian temple
18	5	A wizard's hat	20	10	Aerial view of a promenade
18	7	Reflective pond scene	20	11	Image with holographic retro synthwave aesthetics
18	8	An image of Andorra	20	15	Colorful urban art
18	11	Image with a whirlpool in the ocean	21	0	Timeless black and white
18	12	Photo taken in the Hawaiian volcanoes	21	1	Picture taken in the Scotland countryside
18	14	serene woodland refuge	21	2	A photo of a woman
18	15	Image taken from a distance	21	3	An image of Andorra
19	0	pristine forest scene	21	4	A bamboo
19	1	An image of Fiji	21	6	Playful urban scenes
19	2	Candid wildlife shot	21	7	an image of portsmouth
19	3	Cheerful adolescents	21	8	Inviting coffee shop
19	4	Photo taken in Bora Bora	21	10	Playful siblings
19	5	A photo of a woman	21	11	Photo captured in the Arizona desert
19	6	Dramatic sunset	21	12	Macro botanical photography
19	7	Striking contemporary sculpture	21	14	Photo taken in Beijing, China
19	8	Close-up view	22	2	Intriguing and mysterious forest pathway
19	9	ocean horizon	22	4	Reflective coastal view
19	11	Picture taken in Laos	22	7	Serene winter wonderland
19	12	Pulsating concert light show	22	9	A photo of food
19	13	Illustration of a hidden fantasy oasis	22	12	Serene beach sunset
19	15	Photo taken in Bangkok, Thailand	22	13	Picture taken in the Netherlands
20	0	Picture taken in a zoo or wildlife sanctuary	22	15	Concerned facial expression
20	1	Picture taken in Seychelles	23	0	Image with shattered glass art installation
20	2	An image of a Emergency Medical Technician (EMT)	23	1	Futuristic scientific advancement
20	3	Intrica wood carvingte	23	3	Bustling city waterfront
20	5	Urban nostalgia	23	5	Picture of trees
20	6	Bustling city square	23	6	Picture taken in Fiji
20	7	an image of liechtenstein	23	7	A picture of a bridge
20	8	Photo taken in Borneo	23	15	Evocative horizons

Table 7: TextSpan of ViT-L-14 baseline model with fine-tuning on waterbirds dataset.

<i>Layer</i>	<i>Head</i>	<i>TextSpan</i>	<i>Layer</i>	<i>Head</i>	<i>TextSpan</i>
18	1	Picture with airplanes	20	7	an image of liechtenstein
18	4	Close-up of a textured plant leaf	20	8	Majestic canyon vista
18	7	Reflective and calm lake surface	20	9	Tranquil Asian temple
18	8	an image of liechtenstein	20	10	Reflective pond scene

18	11	Reflections on water	21	1	Picture taken in the Scotland countryside
18	12	Photo taken in the Hawaiian volcanoes	21	3	An image of Andorra
18	13	Image with a trio of friends	21	4	A bamboo
18	14	serene woodland refuge	21	7	an image of liechtenstein
18	15	An image of a Novelist	21	11	Photo captured in the Arizona desert
19	0	pristine forest scene	21	14	Photo taken in Beijing, China
19	1	Quiet rural farmhouse	22	2	Serene waters
19	4	Photo taken in Bora Bora	22	4	peaceful waterfront scene
19	9	ocean horizon	22	7	Serene winter wonderland
19	11	Picture taken in Ecuador	22	12	Serene beach sunset
19	12	Pulsating concert light show	22	13	Picture taken in the Netherlands
19	15	Photo taken in Bangkok, Thailand	22	15	Tranquil mountain refuge
20	0	Picture taken in the English countryside	23	0	Reflective cityscape at twilight
20	1	Picture taken in Seychelles	23	1	Futuristic transport system
20	2	An image of a Emergency Medical Technician (EMT)	23	3	peaceful waterfront scene
20	3	Intrica wood carvingte	23	5	Serene waters
20	6	Bustling city waterfront	23	6	Image captured in the Australian bushlands

Table 8: Performance comparison between baseline, diversity loss regularization, and user-guided ablated model

Model	(0, 0)	(0, 1)	(1, 0)	(1, 1)	Full	improvement
ViT-14-L- (Baseilne)	94.41%	47.76%	70.56%	93.30%	73.49%	0
ViT-14-L + User-guided-ablation-6-layers	91.53%	76.98%	84.42%	89.72%	84.88%	11.39%
Fine-tuned Baseline	99.82%	89.36%	71.96%	96.26%	92.27%	0
Fine-tuned Baseline + Ldiv	99.82%	87.45%	73.52%	96.73%	91.75%	-0.52%
Fine-tuned Baseline + User-guided ablation 1 layer	99.87%	92.37%	76.01%	94.70%	93.73%	1.46%
Fine-tuned Baseline + User-guided ablation 2 layer	99.82%	93.92%	76.95%	94.39%	94.39%	2.12%
Fine-tuned Baseline + User-guided ablation 3 layer	99.82%	94.06%	77.10%	94.39%	94.46%	2.19%
Fine-tuned Baseline + User-guided ablation 4 layer	99.82%	94.15%	76.95%	94.24%	94.46%	2.19%
Fine-tuned Baseline + User-guided ablation 5 layer	99.82%	94.15%	76.95%	94.24%	94.46%	2.19%
Fine-tuned Baseline + User-guided ablation 6 layer	99.82%	94.46%	77.26%	94.08%	94.60%	2.33%

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